

"APTITUDE ASSESSMENT (IT - A)"

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* Indicates required question

Duration = 100 minutes

*

Which concept allows same function name with different parameters?

- ☐ A) Overriding
- ☒ B) Overloading
- ☐ C) Encapsulation
- ☐ D) Inheritance

*

A product is sold at ₹1200 with 20% profit. What is the cost price?

- ☐ A) 900
- ☒ B) 1000
- ☐ C) 1100
- ☐ D) 950



*

Which sorting is best for **nearly sorted arrays**?

- ☐ A) Heap Sort
- ☐ B) Merge Sort
- ☐ C) Insertion Sort
- ☒ D) Selection Sort

*

Bubble sort best case time complexity:

- ☒ A) $O(n)$
- ☐ B) $O(n \log n)$
- ☐ C) $O(n^2)$
- ☐ D) $O(\log n)$

*

What is the LCM of 12 and 18?

- ☐ A) 24
- ☒ B) 36
- ☐ C) 48
- ☐ D) 72



*

Constructor is called when:

- ☐ A) Object is destroyed
- ☒ B) Object is created
- ☐ C) Program ends
- ☐ D) Variable declared

*

Which algorithm uses Divide and Conquer?

- ☐ A) Bubble Sort
- ☒ B) Quick Sort
- ☐ C) Linear Search
- ☐ D) BFS

*

Synonym of "Happy":

- ☐ A) Sad
- ☒ B) Joyful
- ☐ C) Angry
- ☐ D) Weak



*

Which of the following is **NOT** a process state?

- ☐ A) Ready
- ☐ B) Running
- ☐ C) Waiting
- ☒ D) Terminated Suspended

*

If Square : Cube, then Circle : ?

- ☒ A) Sphere
- ☐ B) Cylinder
- ☐ C) Cone
- ☐ D) Cube

*

Which algorithm detects cycles in graph?

- ☒ A) DFS
- ☐ B) BFS
- ☐ C) Sorting
- ☐ D) Hashing



*

Which scheduling algorithm is **preemptive by nature**?

- ☐ A) FCFS
- ☐ B) SJF (Non-preemptive)
- ☐ C) Priority Scheduling (Non-preemptive)
- ☒ D) Round Robin

*

Which access modifier allows access only inside same package?

- ☐ A) public
- ☐ B) private
- ☐ C) protected
- ☒ D) default

*

Graph with no cycles is called:

- ☐ A) DAG
- ☒ B) Tree
- ☐ C) Forest
- ☐ D) Cyclic Graph



*

A car travels at 60 km/h for 2 hours. Distance covered?

- ☐ A) 100 km
- ☐ B) 110 km
- ☒ C) 120 km
- ☐ D) 130 km

*

Output of C code:

```
int a=5;
```

```
printf("%d", a++ + ++a);
```

- ☐ A) 11
- ☐ B) 12
- ☒ C) Undefined
- ☐ D) 10



*

Which protocol is used to map IP address to MAC address?

- ☐ A) DNS
- ☒ B) ARP
- ☐ C) RARP
- ☐ D) ICMP

*

Which SQL command removes **all rows but keeps table structure**?

- ☐ A) DROP TABLE
- ☐ B) DELETE FROM TABLE
- ☒ C) TRUNCATE TABLE
- ☐ D) REMOVE TABLE

*

A complete binary tree with height h has maximum nodes:

- ☐ A) $2h$
- ☒ B) $2^{(h+1)} - 1$
- ☐ C) h^2
- ☐ D) h



*

Worst case time complexity of Merge Sort:

- ☐ A) $O(n)$
- ☒ B) $O(n \log n)$
- ☐ C) $O(n^2)$
- ☐ D) $O(\log n)$

*

Passage: "Education improves knowledge and skills."

Question: What does education improve?

- ☐ A) Wealth
- ☒ B) Knowledge and skills
- ☐ C) Power
- ☐ D) Health

*

Which data structure is used in LRU cache?

- ☐ A) Stack
- ☐ B) Queue
- ☒ C) HashMap + Doubly Linked List
- ☐ D) Tree



*

Which is NOT a linear data structure?

- ☐ A) Stack
- ☐ B) Queue
- ☒ C) Graph
- ☐ D) Linked List

*

2, 4, 8, 16, ?

- ☐ A) 18
- ☐ B) 24
- ☒ C) 32
- ☐ D) 30

*

What is the main purpose of **spooling**?

- ☐ A) Speed up CPU execution
- ☐ B) Improve disk performance
- ☒ C) Overlap I/O and CPU processing
- ☐ D) Increase memory



*

Which OSI layer is responsible for logical addressing and routing?

- ☐ A) Data Link Layer
- ☒ B) Network Layer
- ☐ C) Transport Layer
- ☐ D) Application Layer

*

If profit increased from ₹400 to ₹600, the increase is:

- ☐ A) 25%
- ☐ B) 33.3%
- ☒ C) 50%
- ☐ D) 75%

*

Static members belong to:

- ☐ A) Object
- ☒ B) Class
- ☐ C) Function
- ☐ D) Package



*

Which structure is best for priority scheduling?

- ☐ A) Stack
- ☐ B) Queue
- ☒ C) Heap
- ☐ D) Array

*

Which of the following is a **DDL** command?

- ☐ A) SELECT
- ☐ B) INSERT
- ☐ C) DELETE
- ☒ D) CREATE

*

Which keyword is used to allocate memory dynamically in C++?

- ☐ A) malloc
- ☐ B) calloc
- ☒ C) new
- ☐ D) alloc



*

Which traversal prints root first?

- ☐ A) Inorder
- ☒ B) Preorder
- ☐ C) Postorder
- ☐ D) Level order

*

Which data structure follows **LIFO**?

- ☐ A) Queue
- ☒ B) Stack
- ☐ C) Array
- ☐ D) Tree

*

Which is **NOT** supported in Java?

- ☐ A) Multiple inheritance using classes
- ☐ B) Multiple inheritance using interfaces
- ☐ C) Method overloading
- ☐ D) Method overriding



*

Which normal form removes **transitive dependency**?

- ☐ A) 1NF
- ☐ B) 2NF
- ☐ C) 3NF
- ☐ D) BCNF

*

Which algorithm finds shortest path with negative edges?

- ☐ A) Dijkstra
- ☐ B) Bellman-Ford
- ☐ C) Floyd-Warshall
- ☐ D) BFS

*

Which concept hides internal details and shows only functionality?

- ☐ A) Polymorphism
- ☐ B) Encapsulation
- ☐ C) Abstraction
- ☐ D) Inheritance



*

Output of Java code:

```
int x = 10;
```

```
System.out.println(x++ + ++x);
```

- ☐ A) 20
- ☐ B) 21
- ☐ C) 22
- ☐ D) Undefined

*

```
for(i=1; i<=n; i++)
```

```
    for(j=1; j<=n; j=j*2)
```

Time Complexity is:

- ☐ A) $O(n)$
- ☐ B) $O(n \log n)$
- ☐ C) $O(n^2)$
- ☐ D) $O(\log n)$



*

Binary search time complexity:

- ☐ A) $O(n)$
- ☐ B) $O(\log n)$
- ☐ C) $O(n \log n)$
- ☐ D) $O(n^2)$

*

In Java, default value of instance variable is:

- ☐ A) Garbage
- ☐ B) 0
- ☐ C) Null
- ☐ D) Depends on datatype

*

Which layer of OSI model is responsible for encryption, compression, and translation?

- ☐ A) Session Layer
- ☐ B) Presentation Layer
- ☐ C) Transport Layer
- ☐ D) Physical Layer



*

Which shape completes the pattern:
Circle → Square → Triangle → Circle → ?

- ☐ A) Square
- ☐ B) Triangle
- ☐ C) Rectangle
- ☐ D) Pentagon

*

Which is NOT comparison sorting?

- ☐ A) Quick Sort
- ☐ B) Merge Sort
- ☐ C) Radix Sort
- ☐ D) Heap Sort



*

What is the output?

```
int a=3;
```

```
printf("%d", sizeof(a++));
```

- ☐ A) 3
- ☐ B) 4
- ☐ C) Compiler error
- ☐ D) Depends

*

If CAT = DBU, then DOG = ?

- ☐ A) EPH
- ☐ B) EOG
- ☐ C) EPH
- ☐ D) DOH



*

Stack is mainly used in:

- ☐ A) Graph traversal
- ☐ B) Recursion
- ☐ C) Sorting
- ☐ D) File handling

*

Statement & Conclusion

Statement: All roses are flowers. Some flowers are red.

Conclusion: Some roses are red.

- ☐ A) True
- ☐ B) False
- ☐ C) Both true & false
- ☐ D) Cannot be determined



*

Which is NOT a valid pointer type?

- ☐ A) int *p
- ☐ B) void *p
- ☐ C) char *p
- ☐ D) int &p

*

The ratio of boys to girls in a class is 3:2. If there are 30 students, how many girls?

- ☐ A) 10
- ☐ B) 12
- ☐ C) 18
- ☐ D) 20

*

A can complete a work in 10 days, B in 15 days. Together they can finish in:

- ☐ A) 4 days
- ☐ B) 5 days
- ☐ C) 6 days
- ☐ D) 7 days



*

A complete binary tree with n nodes has height:

- ☐ A) $\log n$
- ☐ B) n
- ☐ C) $n \log n$
- ☐ D) $\sqrt{n}$

*

In paging, logical address consists of:

- ☐ A) Segment number + Offset
- ☐ B) Page number + Offset
- ☐ C) Base address + Limit
- ☐ D) Frame number + Page number

*

Which isolation level allows **dirty reads**?

- ☐ A) Read Committed
- ☐ B) Repeatable Read
- ☐ C) Serializable
- ☐ D) Read Uncommitted



*

Sentence Correction

He **go** to school daily.

- ☐ A) goes
- ☐ B) gone
- ☐ C) going
- ☐ D) go

*

Finding a triangle inside a complex figure is called:

- ☐ A) Series
- ☐ B) Analogy
- ☐ C) Embedded Figure
- ☐ D) Mirror Image



*

Ravi is the son of Sita. Sita is the wife of Ram. How is Ram related to Ravi?

- ☐ A) Brother
- ☐ B) Father
- ☐ C) Uncle
- ☐ D) Grandfather

*

Time complexity of heap insertion:

- ☐ A) $O(1)$
- ☐ B) $O(\log n)$
- ☐ C) $O(n)$
- ☐ D) $O(n \log n)$

*

Time complexity of BFS:

- ☐ A) $O(V+E)$
- ☐ B) $O(V^2)$
- ☐ C) $O(E^2)$
- ☐ D) $O(\log V)$



*

SI on ₹1000 at 10% per annum for 2 years is:

- ☐ A) 100
- ☐ B) 200
- ☐ C) 300
- ☐ D) 400

*

What is the output?

```
int x=5;
```

```
printf("%d", x<<2);
```

- ☐ A) 10
- ☐ B) 20
- ☐ C) 5
- ☐ D) 8



*

Diamond problem occurs in:

- ☐ A) Single inheritance
- ☐ B) Multiple inheritance
- ☐ C) Hierarchical inheritance
- ☐ D) Hybrid inheritance

*

Which traversal uses queue?

- ☐ A) Inorder
- ☐ B) Preorder
- ☐ C) Postorder
- ☐ D) Level Order

*

Which command is used to **add a new column** to an existing table?

- ☐ A) MODIFY TABLE
- ☐ B) ALTER TABLE
- ☐ C) UPDATE TABLE
- ☐ D) INSERT TABLE



*

Which scheduling algorithm is **fair but not efficient for CPU-bound processes**?

- ☐ A) FCFS
- ☐ B) Round Robin
- ☐ C) SJF
- ☐ D) Priority Scheduling

*

If a paper is folded once and cut, how many pieces after unfolding?

- ☐ A) 1
- ☐ B) 2
- ☐ C) 4
- ☐ D) 8

*

In binary search, array must be:

- ☐ A) Random
- ☐ B) Sorted
- ☐ C) Unsorted
- ☐ D) Circular



*

Which key uniquely identifies each record in a table?

- ☐ A) Foreign Key
- ☐ B) Candidate Key
- ☐ C) Primary Key
- ☐ D) Secondary Key

*

Which is NOT a valid loop?

- ☐ A) for
- ☐ B) while
- ☐ C) foreach
- ☐ D) until

*

The average of 5 numbers is 20. If one number is removed, the average becomes 18. What is the removed number?

- ☐ A) 26
- ☐ B) 28
- ☐ C) 30
- ☐ D) 32



*

Which page replacement algorithm **never suffers from Belady's anomaly**?

- ☐ A) FIFO
- ☐ B) LRU
- ☐ C) Optimal
- ☐ D) Random

*

Which algorithm is stable?

- ☐ A) Heap Sort
- ☐ B) Quick Sort
- ☐ C) Merge Sort
- ☐ D) Selection Sort

*

Best sorting for nearly sorted data:

- ☐ A) Selection Sort
- ☐ B) Insertion Sort
- ☐ C) Heap Sort
- ☐ D) Merge Sort



*

If time quantum is too large in Round Robin scheduling, it behaves like:

- ☐ A) SJF
- ☐ B) Priority Scheduling
- ☐ C) FCFS
- ☐ D) Preemptive Scheduling

*

Worst case time complexity of Quick Sort is:

- ☐ A) $O(n \log n)$
- ☐ B) $O(n^2)$
- ☐ C) $O(\log n)$
- ☐ D) $O(n)$

*

A company sales:

2019 = 200, 2020 = 250, 2021 = 300

What is the percentage increase from 2019 to 2021?

- ☐ A) 25%
- ☐ B) 50%
- ☐ C) 75%
- ☐ D) 100%



*

Function overriding is an example of:

- ☐ A) Compile-time polymorphism
- ☐ B) Run-time polymorphism
- ☐ C) Encapsulation
- ☐ D) Abstraction

*

Antonym of "Increase":

- ☐ A) Rise
- ☐ B) Grow
- ☐ C) Decrease
- ☐ D) Add

*

Which data structure is used in **Undo/Redo** operations?

- ☐ A) Queue
- ☐ B) Stack
- ☐ C) Tree
- ☐ D) Graph



*

Which index improves **search performance but slows insert/update?**

- ☐ A) Primary index
- ☐ B) Secondary index
- ☐ C) Clustered index
- ☐ D) All of the above

*

Which traversal uses a queue?

- ☐ A) DFS
- ☐ B) BFS
- ☐ C) Inorder
- ☐ D) Postorder

*

Which device is used to connect two different networks?

- ☐ A) Hub
- ☐ B) Switch
- ☐ C) Router
- ☐ D) Repeater



*

Which of the following is **NOT** a necessary condition for deadlock?

- ☐ A) Mutual Exclusion
- ☐ B) Hold and Wait
- ☐ C) Preemption
- ☐ D) Circular Wait

*

Which operator is used for **bitwise AND**?

- ☐ A) &&
- ☐ B) &
- ☐ C) |
- ☐ D) ^

*

Mirror image of "b" is:

- ☐ A) b
- ☐ B) d
- ☐ C) p
- ☐ D) q



*

Which constraint ensures no duplicate values in a column?

- ☐ A) PRIMARY KEY
- ☐ B) FOREIGN KEY
- ☐ C) UNIQUE
- ☐ D) CHECK

*

Which is NOT an OOPS principle?

- ☐ A) Encapsulation
- ☐ B) Polymorphism
- ☐ C) Inheritance
- ☐ D) Compilation

*

Maximum nodes in binary tree of height h:

- ☐ A) $2h$
- ☐ B) $2^{(h+1)} - 1$
- ☐ C) h^2
- ☐ D) h



*

Five persons A, B, C, D, E are sitting in a row. C is in the middle. A is left of B. Who is in the middle?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

*

Worst case of Quick Sort:

- ☐ A) $O(n)$
- ☐ B) $O(n \log n)$
- ☐ C) $O(n^2)$
- ☐ D) $O(\log n)$

*

A number is increased by 20% and then decreased by 20%. What is the net change?

- ☐ A) 0%
- ☐ B) 4% decrease
- ☐ C) 4% increase
- ☐ D) 2% decrease



*

Which protocol is connectionless and unreliable but faster?

- ☐ A) TCP
- ☐ B) HTTP
- ☐ C) UDP
- ☐ D) FTP

*

Which is stored in **Heap memory**?

- ☐ A) Local variables
- ☐ B) Static variables
- ☐ C) Dynamic objects
- ☐ D) Registers

*

Which is NOT a property of heap?

- ☐ A) Complete binary tree
- ☐ B) Heap property
- ☐ C) Balanced BST
- ☐ D) Root is max/min



*

Thrashing occurs when:

- ☐ A) CPU utilization is high
- ☐ B) Process execution is fast
- ☐ C) System spends more time paging than executing
- ☐ D) Memory is full

*

Which SQL clause is used to **filter aggregated results**?

- ☐ A) WHERE
- ☐ B) GROUP BY
- ☐ C) HAVING
- ☐ D) ORDER BY

*

In which scheduling algorithm **average waiting time is minimum theoretically**?

- ☐ A) FCFS
- ☐ B) SJF
- ☐ C) Round Robin
- ☐ D) Priority Scheduling



*

Arrange:

1. He bought a book
2. He went to the shop
3. He read the book

- ☐ A) 1-2-3
- ☐ B) 2-1-3
- ☐ C) 3-2-1
- ☐ D) 2-3-1

*

Which join returns **only matching records from both tables**?

- ☐ A) LEFT JOIN
- ☐ B) RIGHT JOIN
- ☐ C) FULL JOIN
- ☐ D) INNER JOIN



*

Which keyword prevents method overriding in Java?

- ☐ A) static
- ☐ B) final
- ☐ C) private
- ☐ D) protected

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